TOP 50 CASE STUDY QUESTIONS

Applied Mathematics Class 12 | CBSE Board Pattern

Curated by Thunderstudy

🟣 PART 1 — LINEAR PROGRAMMING (High Weightage ⭐)

Case Study 1: Factory Production

A factory produces chairs and tables. Profit per chair = ₹40, per table = ₹60. Labour ≤ 100 hrs, Wood ≤ 80 units.

1. Write objective function. | 2. Write constraints. | 3. Identify feasible region. | 4. Where is maximum profit obtained?

Case Study 2: Diet Problem

A diet requires protein and vitamins from foods A & B.

1. Form inequalities. | 2. Plot feasible region. | 3. Identify corner points. | 4. Find optimum solution rule.

Case Study 3: Transport Company

Two trucks carry goods with fuel and weight limits.

1. Define decision variables. | 2. Write constraints. | 3. Graphical representation meaning. | 4. Why optimum occurs at vertices?

Case Study 4: Bakery Production

Bread & cakes using flour and sugar limits.

1. Objective function | 2. Feasible region meaning | 3. Corner point importance | 4. Maximum profit concept

Case Study 5: Furniture Workshop

Tables & desks manufacturing constraints.

1. Linear inequality formation | 2. Non-negativity restriction | 3. Feasible solution meaning | 4. Optimal solution

Cases 6–10 (Recurring Themes): Garment factory planning, Advertising budget allocation, Milk distribution optimization, School resource allocation, Chemical mixture optimization.

🟣 PART 2 — FINANCIAL MATHEMATICS

Case Study 11: EMI Loan

Loan ₹5,00,000 at 7.5%.

1. Meaning of EMI | 2. Flat vs reducing rate | 3. Interest component | 4. EMI dependence on time

Case Study 12: Depreciation of Machine

Machine cost ₹1,00,000, scrap ₹5,000.

1. Depreciation meaning | 2. Book value concept | 3. Linear depreciation | 4. Economic interpretation

Case Study 13: Investment Growth

Investment grows annually.

1. CAGR formula | 2. Growth interpretation | 3. Compounding effect | 4. Long-term investment advantage

Case Study 14: Sinking Fund

Company saves yearly for replacement.

1. Purpose of sinking fund | 2. Future value relation | 3. Interest effect | 4. Time dependency

Cases 15–18: Insurance premium calculation, Retirement planning annuity, Recurring deposit analysis, Effective vs nominal interest.

🟣 PART 3 — PROBABILITY & DISTRIBUTION

Case Study 19: Coin Toss Experiment

1. Random variable | 2. Expectation | 3. Probability distribution | 4. Mean interpretation

Case Study 20: Quality Control

Defective items probability in production.

1. Binomial condition | 2. Mean & variance | 3. Success probability | 4. Real-life meaning

Case Study 21: Call Center Data (Poisson)

Calls per minute recorded over a shift.

1. Poisson assumption | 2. Mean = variance | 3. Probability event | 4. Application area

Case Study 22: Normal Distribution (Heights)

Student heights follow a bell-shaped curve.

1. Symmetry property | 2. Mean=Median=Mode | 3. SD meaning | 4. Bell shape interpretation

Cases 23–25: Lottery probability, Dice experiment expectation, Customer arrival analysis.

🟣 PART 4 — STATISTICS & TIME SERIES

Case Study 26: Sales Trend Analysis

1. Trend component | 2. Seasonal variation | 3. Moving average purpose | 4. Forecast meaning

Case Study 27: Weather Data

Temperature changes recorded monthly.

1. Seasonal variation ID | 2. Cyclical vs irregular | 3. Time series components | 4. Prediction method

Cases 28–30: Agricultural production data, Tourism growth analysis, Online sales forecasting.

🟣 PART 5 — CALCULUS IN ECONOMICS

Case Study 31: Cost Function C(x)

1. Marginal cost | 2. Average cost | 3. Minimum cost condition | 4. Economic meaning

Case Study 32: Revenue Function

1. Revenue formula | 2. MR derivation | 3. Profit maximization | 4. MR = MC rule

Case Study 33: Population Growth

1. Differential equation form | 2. Growth constant | 3. Exponential model | 4. Interpretation

Cases 34–36: Inventory cost optimization, Production rate analysis, Advertising revenue model.

🟣 PART 6 — MATRICES & APPLICATIONS

Case Study 37: Transportation Matrix

1. Matrix representation | 2. Order of matrix | 3. Multiplication meaning | 4. Practical interpretation

Case Study 38: Input-Output Model

1. Matrix usage | 2. Economic balance | 3. Solution meaning | 4. Real-life significance

Cases 39–40: Encryption matrix model, Population transition matrix.

🟣 PART 7 — ASSERTION-REASON STYLE CASES

Cases 41–50: Behaviour of increasing functions, Inequality graph interpretation, Feasible region properties, Objective function equality, Hypothesis testing, Degrees of freedom, Variance properties, Probability limits, Continuity, Optimization in business.

⭐ BOARDS FAVOURITE REPEATED TOPICS

- **Linear Programming:** Always appears in Section E.
- **Financial Maths:** Focusing on EMI and Depreciation.
- **Normal Distribution:** Characteristics of the bell curve.
- **Time Series:** Forecasting using trends.
- **Calculus:** Cost-Revenue optimization (MR=MC).

Top 50 Case Studies By Thunderstudy | For Class 12 Boards Revision